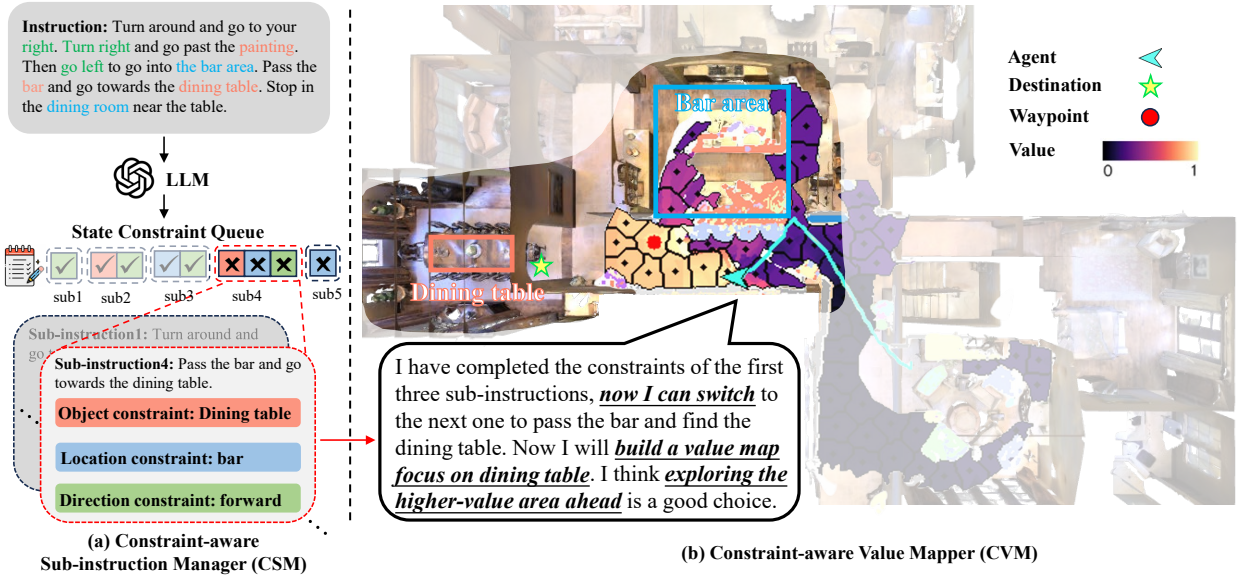

Instruction: Turn around and go to your right. Turn right and go past the painting. Then go left to go into the bar area. Pass the bar and go towards the dining table. Stop in the dining room near the table.
LLM
State Constraint Queue
sub1
sub2
sub3
sub4
sub5
Sub-instruction1: Turn around and
Sub-instruction4: Pass the bar and go towards the dining table.
Object constraint: Dining table
Location constraint: bar
Direction constraint: forward
(a) Constraint-aware Sub-instruction Manager (CSM)
Bar area
Dining table
Agent
Destination
Waypoint
Value
0
1
I have completed the constraints of the first three sub-instructions, now I can switch to the next one to pass the bar and find the dining table. Now I will build a value map focus on dining table. I think exploring the higher-value area ahead is a good choice.
(b) Constraint-aware Value Mapper (CVM)